LAB-7

Principal Component Analysis and Tasseled Cap transformation

Use dc image for the Task 1
Use tm-Atlanta image for the Task 2

Task-1: Perform Principal Component Analysis

Go to **Raster |Resolution |Spectral |Principal component**.

Principal component dialog box opens, select input file and name the output file, select 6 in the number of components required and click ok.

Answer following Questions:

1. Analyze different components in gray scale image and explain how much information in each component?
2. Which component has highest information?
3. What type of information in fourth component provides and what it is, explain?
4. Which components would you prefer to use for further analysis? Make a color composite image of three preferred components and analyze the information contents as compared to gray scale components?

Task-2: Perform Tasseled Cap Transformation

Go to **Raster |Resolution |Spectral | Tasseled Cap**

Tasseled Cap dialog box opens, select input file and name the output file, create four components, click ok.

Answer following Questions:

1. Analyze different Tasseled Cap components, such as brightness, greenness, yellowness, wetness /nonesuch layer?
2. Which two tasseled components have maximum information of vegetation?
3. What is the difference between Principal Component Analysis and Tasseled Cap transformation for different types of surface features?